Machine-checked degree lower bounds for interpolating the elliptic curve discrete logarithm from x-coordinates

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Abstract

We record three degree lower bounds for polynomials that interpolate functions of the discrete logarithm from the x-coordinates of multiples kP of a point on a short Weierstrass curve, and we give Lean 4 formalizations of them. (B) If $g(x(kP)) = k^2$ for $1 \le k \le (r-1)/2$, where P has odd order r on a short Weierstrass curve over a field F with $2 \neq 0$ and $3 \neq 0$, then $r \le 6 \deg g + 5$. (C) If $g(x(kP)) = k$ for $1 \le k \le N$ and $2N < r$, then $N \le 3 \deg g + 2$. (D) If the interpolation $g(x(kP)) = k^2$ is required only on a set $A \subseteq [1, N]$, $2N < r$, then for each $j \ge 1$ at most $3 \deg g$ elements $k \in A$ have $k-j, k+j \in A$ (under an explicit extra hypothesis). All three are proved by one device, a cleared second-difference polynomial of exact degree $3 \deg g$ (constant g is treated separately). The statements are verified by Lean against a pinned Mathlib; the Lean statements are quoted verbatim. Interpolation degree bounds of this kind are due to Coppersmith–Shparlinski, Lange–Winterhof and Satoh; we could not access their exact statements, so our bounds may be known or weaker. The contribution claimed is the machine-checked formalization and the precise hypotheses. The results concern polynomial degree in a restricted model and have no bit-complexity consequence. A clearly marked later part (Section 7) records, with their evidence levels kept separate, six further items: three formally verified barriers for restricted algebraic models (division-free straight-line programs; straight-line programs with division; computation trees with equality tests, Theorem DT, stated with its scope), a method barrier (a proof attempt, not reviewed), an experimental conjecture, and a proof-reviewed but *not formalized* bound (Theorem E). Section 8 tabulates the laboratory's benchmarked algorithm candidates, all of which were negative. **No improvement over Pollard rho is claimed anywhere in this report.**

Evidence levels used in this report. We keep four levels apart and name the level of every item. *Formally verified (Lean)*: a Lean 4 declaration checked against the pinned Mathlib, with the statement quoted. *Reviewed proof*: a paper proof that an independent AI reviewer examined, not formalized. *Experiment*: a measurement at toy sizes; it proves nothing about large parameters. *Conjecture*: an unproved statement. Review by AI agents of the same provider is only partially independent; no human has verified anything here.

1 Introduction

The elliptic curve discrete logarithm problem (ECDLP) asks, given P and $Q = kP$ in a group of order r, for k modulo r. A natural question about restricted classes of algorithms is whether the map $x(kP) \mapsto k$ (or a quantity such as k^2, which is the natural x-only invariant because $x(kP) = x(-kP)$) can be given by a low-degree polynomial. Here k is a positive integer representative with $k \le (r-1)/2$, and the target k^2 (or k) is its image in the field F; it is not a function of k modulo r in general (over $\mathbb{F}_p$ with $p \ge 5$ and $k \le (r-1)/2 < p$ no wrap-around occurs,

but over extension fields k may exceed the characteristic). A lower bound on the degree of any interpolating polynomial shows that no *low-degree-polynomial formula* computes the logarithm; it is a barrier for one restricted model, in the spirit of the interpolation results cited in Section 6.

What such a bound does *not* say should be stated plainly. It says nothing about algorithms that use other operations (comparisons, branching, group operations, index calculus), nothing about lower bounds on the cost of evaluating it beyond the logarithmic one noted next, and nothing about bit complexity. Degree and evaluation cost are related only weakly: Horner's rule evaluates a degree-d polynomial with at most d multiplications, so degree is an upper bound for cost in that model; conversely a division-free straight-line program with s multiplications computes a polynomial of degree at most 2^s, so a degree lower bound yields only a *logarithmic* lower bound on the number of multiplications (see Section 7 for the formalized form), and no statement about bit operations. A high-degree polynomial such as X^{p-2} can be cheap to evaluate. The open target of the laboratory (a universal classical ECDLP algorithm with worst-case $O(n \log n)$ bit operations) is unaffected by this report.

Contents. Section 2 fixes notation; Section 3 states the three results; Section 4 gives the proofs; Section 5 quotes the Lean statements and the trust base; Section 6 discusses prior work and novelty; Section 7 is a marked part with six later laboratory items (i)–(vi) (it contains the division-SLP barrier, Theorem DT and Theorem E); Section 8 tabulates benchmarked algorithm candidates; Section 9 lists limitations.

Existence is not at issue: the nodes $x(kP)$, $1 \le k \le N$, $2N < r$, are pairwise distinct (Section 2), so Lagrange interpolation always gives g with $\deg g \le N-1$ for the prescribed values; for Theorem B this is $\deg g \le (r-3)/2$. Theorems B and C therefore determine the minimal interpolation degree on a full range up to a constant factor (about 3), and the bounds are linear in r. Theorem D is a partial-agreement count, not a minimal-degree statement on a range, and is not covered by this sentence.

2 Setting and notation

Let F be a field and $W : y^2 = x^3 + ax + b$ a short Weierstrass cubic over F ($a_1 = a_2 = a_3 = 0$, $a = a_4$, $b = a_6$). The principal case is $F = \mathbb{F}_p$ and W an elliptic curve with a point P of prime order r. For a nonzero affine point Q we write $x(Q)$ for its x-coordinate, and we write $r = \mathrm{ord}(P)$ for the additive order of P. For a polynomial $g \in F[X]$ we write $m = \deg g$ and $\mathrm{lc}(g)$ for its leading coefficient. Throughout we assume $2 \neq 0$ in F (written $2 \neq 0$; Theorem B also uses $3 \neq 0$).

Two facts are used. First, if $2N < r$ then $k \mapsto x(kP)$ is injective on $\{1, \dots, N\}$: $x(kP) = x(lP)$ forces $kP = \pm lP$, i.e. $k \equiv \pm l \pmod r$, and $2 \le k + l \le 2N < r$ rules out $k \equiv -l$, so $k \equiv l$, hence $k = l$. Second, the x-addition identities: for $Q = (u, v)$, $P = (x_0, y_0)$ affine points with $u \neq x_0$,

$$\big(x(Q+P) + x(Q-P)\big)(u - x_0)^2 = 2\big((x_0 u + a)(u + x_0) + 2b\big), \tag{1}$$

$$x(Q+P)\, x(Q-P)\, (u - x_0)^2 = (x_0 u - a)^2 - 4b(u + x_0). \tag{2}$$

For (1) write $s_\pm = x(Q \pm P) = \big((v \mp y_0)/(u - x_0)\big)^2 - (u + x_0)$ and use $v^2 + y_0^2 = u^3 + x_0^3 + a(u + x_0) + 2b$ together with $(u + x_0)(u - x_0)^2 = u^3 - u^2 x_0 - u x_0^2 + x_0^3$; (2) is a similar computation (it is part of the formalized algebraic core, Section 5).

3 Main results

Theorem B (k^2-interpolation)**.** *Let W be a short Weierstrass curve over a field F with $2 \neq 0$ and $3 \neq 0$ in F, and P a point of odd order r. If $g \in F[X]$ satisfies $g(x(kP)) = k^2$ for $1 \le k \le (r-1)/2$, then $r \le 6 \deg g + 5$; that is,* $\deg g \ge (r-5)/6$.

For an elliptic curve over $\mathbb{F}_p$ and P of prime order $r > 2$ this is the statement recorded as HYP-eda6732a (the ledger statement assumes r prime; the blind review REV-83ec2fc9 notes that "r prime" can be weakened to "r odd", which is the formalized form). The hypothesis $3 \neq 0$ is formalized but, according to REV-15104e72, not needed mathematically (see Section 4).

Theorem C (linear interpolation)**.** *Let W be short Weierstrass over F with $2 \neq 0$, P a point with $\mathrm{ord}(P) = r$, and $2N < r$. If $g \in F[X]$ satisfies $g(x(kP)) = k$ in F for $1 \leq k \leq N$, then $N \leq 3 \deg g + 2$. In particular, for $N = (r-1)/2$ (r odd), $r \leq 6 \deg g + 5$.*

(HYP-42f91290.) The assumption $3 \neq 0$ is not needed.

Theorem D (partial agreement)**.** *Let W be short Weierstrass over F with $2 \neq 0$, $\mathrm{ord}(P) = r$, $2N < r$ (for r odd, equivalently $N \leq (r-1)/2$), $A \subseteq \{1, \ldots, N\}$, and $g \in F[X]$ with $g(x(kP)) = k^2$ for all $k \in A$. For $j \geq 1$ let $S_j = \#\{k \in A : k - j \in A,\ k + j \in A\}$ (the k with $j < k$ automatically). If $\deg g \geq 1$ or $j \neq 0$ in F, then $S_j \leq 3 \deg g$.*

(HYP-a1dd5501, part (2); the main inequality only.) The side condition is needed over general fields. An instance found by the adversarial review REV-15104e72 (checked there by computer, not by us and not in Lean): over $\mathbb{F}_{27} = \mathbb{F}_3[t]/(t^3 - t + 1)$ the curve $y^2 = x^3 + t^2 x$ has a point P of order 28; with $N = 7$ (so $2N = 14 < 28$), $A = \{1, 4, 7\}$, $g = 1$ and $j = 3$ one has $g(x(kP)) = k^2$ on A ($k^2 \in \{1, 16, 49\} = \{1\}$ in characteristic 3), $\deg g = 0$ and $S_3 = 1 > 0$. This instance needs $r > 4p + 2$, which is impossible over $\mathbb{F}_p$ by the Hasse argument below; that argument does not extend to $\mathbb{F}_q$. Over $\mathbb{F}_p$ the condition excludes only constant g with $p \mid j$, and then $S_j = 0$ for the following reason (*not* formalized in Lean): $S_j \geq 1$ requires $k - j \geq 1$ and $k + j \leq N$, so $N \geq 2j + 1 \geq 2p + 1$, whence $r > 2N \geq 4p + 2$, contradicting the Hasse bound $r \leq \#E(\mathbb{F}_p) \leq p + 1 + 2\sqrt{p}$. Hence over $\mathbb{F}_p$ the unconditional informal statement $S_j \leq 3 \deg g$ holds. (One must not argue $j \leq N < p$; that is false in general, e.g. $y^2 = x^3 + 2x + 1$ over $\mathbb{F}_3$ with $r = 7$, $N = 3$.)

Remark 1 (Hypotheses and the refuted transcription). The hypothesis $2N < r$ in Theorem D cannot be dropped. The ledger record HYP-5da6025e (status: refuted) contains a transcription of part (2) with $N < \min(p, r)$ in place of $N \leq (r-1)/2$; the blind review REV-ac47cc67 produced the counterexample $y^2 = x^3 + 5$ over $\mathbb{F}_7$, $P = (3, 2)$ of order $7 = p$, $A = \{1, \ldots, 6\}$, $g = 5X$, which satisfies $g(x(kP)) = k^2$ for $k = 1, \ldots, 6$ with $\deg g = 1 < S_1/3 = 4/3$. The hypothesis $N \leq (r-1)/2$ had been present in the original agent report and was lost in a transcription error by the principal investigator; the corrected statement is HYP-a1dd5501 and is what is formalized here. Ledger parts of HYP-a1dd5501 not covered here: part (1) for rational functions on E (reviewed, not formalized), and the density corollary $\deg g \geq ((3\delta - 2)N - 2)/3$ for $\delta > 2/3$ (reviewed, not formalized).

Remark 2 (Background: Theorem A). The ledger also records (HYP-adb581ff, status *open*, reviewed, not formalized) a result for rational functions f on E with $f(kP) = k$ for $1 \leq k \leq N \leq r - 1$: the pole-divisor degree is at least $(N-1)/2$, and polynomial total degree at least $(N-1)/6$. It is recorded as known in kind (Section 6) and is cited only as background.

4 Proofs

Assume $m = \deg g \geq 1$ for the moment; constant g is not covered by the exact-degree argument and is treated separately at the end of each case. Fix an affine point $P_0 = (x_0, y_0)$ ($P_0 = P$ in Theorems B, C; $P_0 = jP$ in Theorem D) and put $d = u - x_0$.

The cleared second-difference polynomial. For a variable u let $s_\pm$ be the (formal) roots of $z^2 - e_1 z + e_2$ where $e_1 = s_+ + s_-$ and $e_2 = s_+ s_-$ are the rational functions in u given by (1) and (2) (so $g(s_+) + g(s_-)$ is *defined* as the symmetric expression $\sum_i c_i p_i(e_1, e_2)$ below,

and is evaluated at $u = x(Q)$ only after clearing denominators): $e_1 = n_1/d^2$, $e_2 = n_2/d^2$ with n_1, n_2 polynomials of degree ≤ 2. Write $g = \sum_{i \le m} c_i X^i$. Then $g(s_+) + g(s_-) = \sum_i c_i p_i$ with $p_i = s_+^i + s_-^i$ a polynomial with integer coefficients in e_1, e_2 whose monomials $e_1^\alpha e_2^\beta$ satisfy $\alpha + 2\beta = i$ (Newton–Girard). Multiplying a monomial by d^{2m} gives $n_1^\alpha n_2^\beta d^{2(m-\alpha-\beta)}$, a polynomial of degree $\le 2(\alpha+\beta) + 2(m-\alpha-\beta) = 2m$, because $\alpha + \beta \le i \le m$. Define

$$G_c(u) = d^{2m}\big[g(s_+) + g(s_-)\big] - 2d^{2m} g(u) - c\, d^{2m} \in F[u],$$

for a constant c. The first term has degree $\le 2m$, the third degree $\le 2m$ (and 0 when $c = 0$), and $-2d^{2m} g(u)$ has degree exactly $3m$ with leading coefficient $-2\,\mathrm{lc}(g) \neq 0$ (here $2 \neq 0$ is used). Since $2m < 3m$ for $m \ge 1$:

$$\deg G_c = 3m \quad \text{for every constant } c, \quad \text{in particular } G_c \neq 0. \tag{3}$$

Roots. Suppose $Q = kP$ is affine with $x(Q) \neq x_0$ and $Q \pm P_0$ affine. Then evaluating at $u = x(Q)$, the quantities $s_\pm$ are $x(Q \pm P_0)$ by the identities of Section 2, so $G_c(x(Q)) = d^{2m}\,[g(x(Q+P_0)) + g(x(Q-P_0)) - 2g(x(Q)) - c]$ with $d \neq 0$.

Theorem B. Take $P_0 = P$, $c = 2$. For $2 \le k \le (r-3)/2$ we have $1 \le k-1$ and $k+1 \le (r-1)/2$, so all three values of g are prescribed and the bracket equals $(k+1)^2 + (k-1)^2 - 2k^2 - 2 = 0$. The values $x(kP)$, $2 \le k \le (r-3)/2$, are pairwise distinct (injectivity) and differ from $x(P)$ (since $k \ge 2$), so G_2 has at least $(r-5)/2$ distinct roots. By (3), $(r-5)/2 \le 3m$, i.e. $r \le 6m+5$. Constant $g = c_0$: if $r \le 5$ the bound is trivial; if $r \ge 7$ then $(r-1)/2 \ge 3$ and $g(x(P)) = 1$, $g(x(2P)) = 4$ force $3 = 0$, which $3 \neq 0$ excludes. (According to REV-15104e72, $3 \neq 0$ is superfluous: in characteristic 3 and $r \ge 7$, $k = 3$ gives $9 = 0 \neq 1 = g$. We keep it because it is the formalized hypothesis.) Hence the bound holds.

Theorem C. Take $P_0 = P$, $c = 0$. For $2 \le k \le N-1$ the bracket is $(k+1) + (k-1) - 2k = 0$, giving $N-2$ distinct roots (injectivity as $2N < r$), so $N - 2 \le 3m$. If g is constant and $N \ge 2$, then $g(x(P)) = 1 \neq 2 = g(x(2P))$, contradiction; so $N \le 1 \le 3m + 2$.

Theorem D. Take $P_0 = jP$ (note $j < k \le N < r$). For $k \in C_j = \{k \in A : j < k,\ k \pm j \in A\}$ the bracket with $c = 2j^2$ is $(k+j)^2 + (k-j)^2 - 2k^2 - 2j^2 = 0$. The values $x(kP)$ are distinct for distinct $k \in C_j$ and differ from $x(jP)$ because $k \neq j$ and $k + j \le 2N < r$, so $|C_j| \le 3m$ by (3). If $g = c_0$ is constant, the bracket is $-2j^2 \neq 0$ whenever $j \neq 0$ in F, so G has no roots and $C_j = \emptyset$. □

Exact degree. The computation above shows that, for the second-difference identity, the cleared polynomial has degree exactly $3 \deg g$ and leading coefficient $-2\,\mathrm{lc}(g)$; for Theorem C this is recorded in HYP-42f91290 and agrees with the numerical checks of the ledger.

5 Formal verification

Environment. Toolchain `leanprover/lean4:v4.35.0` (file `lean-toolchain`); Mathlib revision `56a005855ea17e0b0958f73ec801ae140b0a1679` (`lakefile.toml`, with `-DwarningAsError=true`). Source commits recorded in the proof records: Theorem B, PRF-34538f75, commit `cc8dc791a0bd74f89b6cd614898a5773abdc500a` (`lake build`: 3163 jobs, 0 warnings); Theorem C, PRF-35d1387a, commit `8191c0f69398b632fc2c7c7efbfdcdb9e2634b7b` (3164 jobs, 0 warnings); Theorem D, PRF-5357b698, commit `84ddbad82b2f1412e851cf1ab6012b6650bba544` (3165 jobs, 0 warnings). Files: `MathResearch/ECDLP/Interpolation.lean` (generic polynomial core), `CurveX.lean`, `LinearInterp.lean`, `PartialAgreement.lean`. All three proofs use Mathlib's `WeierstrassCurve.Affine.Point`; the curve-specific input is the x-addition identities (`CurveX.sum_prod_identity`, `CurveX.smul_relations`) and injectivity of $k \mapsto x(kP)$.

Statements (verbatim from the repository). Here `xCoord` is the x-coordinate function on `W.Point`, defined to be 0 at the point at infinity (harmless here, since every point evaluated is kP with $1 \le k < r$, which is affine). The binders `{F : Type*} [Field F] [DecidableEq F]` come from `variable` lines of the files; `Affine` is `WeierstrassCurve.Affine` (via `open WeierstrassCurve`) and `F[X]` is `Polynomial F`.

The declarations live in the Lean namespace `MathResearch.ECDLP.CurveX`, in these files: Theorem B in `CurveX.lean`, Theorem C in `LinearInterp.lean`, Theorem D in `PartialAgreement.lean`.

```
theorem degree_bound_short_weierstrass {W : Affine F} (h1 : W.a₁ = 0) (h2 : W.a₂ = 0)
    (h3 : W.a₃ = 0) (h2' : (2 : F) ≠ 0) (h3' : (3 : F) ≠ 0) (P : W.Point) (r : ℕ)
    (hr : addOrderOf P = r) (hodd : r % 2 = 1) (g : F[X])
    (hg : ∀ k : ℕ, 1 ≤ k → k ≤ (r - 1) / 2 → g.eval (xCoord (k • P)) = (k : F) ^ 2)
    :
    r ≤ 6 * g.natDegree + 5
```

```
theorem degree_bound_linear_short_weierstrass {W : Affine F} (h1 : W.a₁ = 0) (h2 : W.a₂
    = 0)
    (h3 : W.a₃ = 0) (h2' : (2 : F) ≠ 0) (P : W.Point) (r : ℕ)
    (hr : addOrderOf P = r) (N : ℕ) (hN : 2 * N < r) (g : Polynomial F)
    (hg : ∀ k : ℕ, 1 ≤ k → k ≤ N → g.eval (xCoord (k • P)) = (k : F)) :
    N ≤ 3 * g.natDegree + 2
```

```
theorem partial_agreement_degree_bound {W : Affine F} (h1 : W.a₁ = 0) (h2 : W.a₂ = 0)
    (h3 : W.a₃ = 0) (h2' : (2 : F) ≠ 0) (P : W.Point) (r : ℕ) (hr : addOrderOf P = r)
    (N : ℕ) (hN : 2 * N < r) (A : Finset ℕ) (hA : ∀ k ∈ A, 1 ≤ k ∧ k ≤ N) (g : F[X])
    (hg : ∀ k ∈ A, g.eval (xCoord (k • P)) = (k : F) ^ 2) (j : ℕ) (hj : 1 ≤ j)
    (hjg : 1 ≤ g.natDegree ∨ (j : F) ≠ 0) :
    (A.filter (fun k => j < k ∧ k - j ∈ A ∧ k + j ∈ A)).card ≤ 3 * g.natDegree
```

The full names are `MathResearch.ECDLP.CurveX.degree_bound_short_weierstrass`, `MathResearch.ECDLP.CurveX.degree_bound_linear_short_weierstrass` and `MathResearch.ECDLP.CurveX.partial_agreement_degree_bound`.

Correspondence to the informal statements. (i) The formal statements are for an arbitrary field F, not only $\mathbb{F}_p$, and for any point P of the given additive order; r need not be prime. Theorem B needs r odd, $2 \neq 0$ and $3 \neq 0$; Theorems C and D need neither $3 \neq 0$ nor r odd (the condition $2N < r$ is equivalent to $N \le (r-1)/2$ for r odd). (ii) Since `W : Affine F` is not assumed nonsingular, the review REV-1f44f52b notes that the results also cover nonsingular points of singular cubics. (iii) Theorem D carries the extra hypothesis `hjg`; the Hasse-bound argument that removes it over $\mathbb{F}_p$ (Section 3) is *not* formalized. (iv) The formal statements are degree bounds only; they assert no existence of g and no complexity. REV-1f44f52b and REV-6c9bfbc1 accepted the correspondence of B and C; the independent review REV-685408b5 (formal correctness only; no complexity or novelty verification) verified the correspondence for D.

Trust base. `#print axioms` on each of the three declarations reports exactly `[propext, Classical.choice, Quot.sound]`. The evidence is per theorem (we verified that each file contains the corresponding `#print axioms` line): Theorem B, `research/ecdlp/runs/RUN-62dbda7c-fe4f-4432-84a2-2068cd516355-command-b585f279-fd16-4cca-b3a1-c2484d545078.evidence` (`degree_bound_short_weierstrass`, together with `sum_prod_identity` and `smul_relations`); Theorem C, `research/ecdlp/runs/RUN-62dbda7c-fe4f-4432-84a2-2068cd516355-command-5da55f17-bbac-4f82-9271-da5f30d66b80.evidence` (`degree_bound_linear_short_weierstrass`); Theorem D, `research/ecdlp/runs/RUN-62db`

`da7c-fe4f-4432-84a2-2068cd516355-command-ca0a1a26-2ee3-4da0-be6c-96a56ffa6cc6.evidence` (`partial_agreement_degree_bound`). Trust therefore rests on the Lean kernel, Mathlib at the pinned revision, and the faithfulness of the statements to the informal ones (reviewed by AI agents of the same provider; independence is partial).

Reproduction. At a checkout of the repository: `lake build` (pinned toolchain and Mathlib), followed by `#print axioms MathResearch.ECDLP.CurveX.degree_bound_short_weierstrass` and likewise for the other two names (see `research/ecdlp/proofs/checks/Round8Axioms.lean` for the pattern).

Empirical sanity checks (not part of the proof). The ledger records toy-size numerical checks: the x-addition identity 2519 times with no failure (Theorem B record); 208 toy cases without violation (REV-83ec2fc9); for Theorem C 533 curves with $p \leq 211$, then 12,459 cases including extension fields (REV-6c9bfbc1); 1242 toy points for B (REV-1f44f52b); 739,847 pairs (A, j) for D (a weak check). These tests do not constitute proofs and are not needed given the Lean proofs; they indicate the bounds are far from tight in tests.

6 Relation to prior work and novelty status

Coppersmith and Shparlinski [1] studied polynomial approximation of the discrete logarithm and the Diffie–Hellman mapping (title of [1]); we did not access the text. Lange and Winterhof (COCOON 2002 [2]; AAECC-15 2003 [3]) and Satoh (WCC 2005, published 2006 [4]) prove degree lower bounds for polynomial interpolation of the elliptic curve discrete logarithm and related maps. **The exact statements of these results were not accessible to the laboratory**: only bibliographic records and abstracts were seen, and full texts were not accessed. We therefore cannot compare constants or hypotheses. Our bounds may be known or weaker. Theorem A is recorded in the ledger as known in kind; the laboratory's novelty checks (ECDLP-22) found no result of the form of Theorems B–D in the accessible material, but this is absence of evidence, not a novelty claim. The laboratory's ledger record for Theorem C (HYP-42f91290) says it is "likely implied by / weaker than Lange–Winterhof-type results" and is *not* claimed novel. Novelty of all three is *unsettled*.

Claimed contribution: a machine-checked formalization (Lean 4, Mathlib, standard axioms only) of the three degree bounds, with explicit, reviewed hypotheses (including the corrected hypothesis $2N < r$ of Theorem D and its generality over arbitrary fields). Bibliographic metadata (first names, venue, pages, year, DOI) of the cited entries were checked against Crossref records (by REV-2b383b9b for [2], and in this revision for [3], [4] and [1]); in particular the year of [4] is 2006. Full texts were still not accessed.

7 Later laboratory results (not part of Theorems B–D)

This section was added in revision 1 and extended in revision 2. It is clearly separate from the results above. Its six items (i)–(vi) carry the statuses stated in the headings; the statuses are those recorded in the laboratory ledger and are not upgraded here.

(i) Division-free straight-line programs: formally verified; independently reviewed for formal correctness. Ledger record PRF-58cf340e (derived from HYP-92a9b4d9; status *formally verified*; the independent batched review REV-7f70c161 (decision *accept*, formal correctness only) targets it. Revision 1 of this report said this review was pending; the ledger now records it, so the statement is updated here). In `MathResearch/ECDLP/SLPBound.lean` a *program* over a commutative ring F is a list of operations with a single input variable x; each operation appends

an arbitrary constant of F, the input, or the sum, difference or product of two earlier values; every operation, including `const` and `var`, counts one step toward the length L (the Lean docstring calls constants "free" only in the sense that any element of F may appear; the `const` operation itself is counted, as REV-7f70c161 also records). There are no divisions, comparisons or branching. A program of length L computes a polynomial of degree at most 2^L in x. Combined with Theorem C this gives, in Lean (source commit `bff5e14c0f187719fb79c07f7768a2856f20604a`, `lake build` 3166 jobs, 0 warnings; build evidence `research/ecdlp/runs/RUN-62dbda7c-fe4f-4432-84a2-2068cd516355-command-94bd6bd6-9c44-47e7-9743-cd567e803d9f.evidence`; axioms `[propext, Classical.choice, Quot.sound]` in `research/ecdlp/runs/RUN-62dbda7c-fe4f-4432-84a2-2068cd516355-command-aeddfabd-5935-4ad5-9911-2c48a87063dc.evidence`):

```
theorem length_bound_linear_short_weierstrass {W : Affine F} (h1 : W.a₁ = 0) (h2 : W.a₂
    = 0)
    (h3 : W.a₃ = 0) (h2' : (2 : F) ≠ 0) (P : W.Point) (r : ℕ) (hr : addOrderOf P = r)
    (N : ℕ) (hN : 2 * N < r) (prog : List (Op F)) (i : ℕ)
    (hprog : ∀ k : ℕ, 1 ≤ k → k ≤ N → (evalAt prog (xCoord (k • P))).getD i 0 = (k
    : F)) :
    N ≤ 3 * 2 ^ prog.length + 2
```

(namespace `MathResearch.ECDLP.SLPBound`; `evalAt` evaluates the program on the input, and out-of-range indices read 0.) Thus a program of length L that outputs k at $x(kP)$ for $1 \le k \le N$, $2N < r$, needs $L \ge \log_2((N-2)/3)$. This answers directly the distinction between polynomial degree and program length raised in review REV-15104e72 (N1): the bound is on the number of ring operations in this restricted model, logarithmic in N. The cost-model consequence recorded in the ledger (if each field operation costs $\Omega(\log p)$ bit operations, then $\Omega(n^2)$ bit operations are needed in this model) is a remark, *not formalized*, and concerns only this restricted model; it says nothing about the open $O(n \log n)$ target for general algorithms.

(ii) Straight-line programs with division: formally verified; independently reviewed for formal correctness. Ledger records HYP-6aef8c30 (status *completed*, outcome *verified*, proof status *formally verified*) and PRF-9586acb8; the independent batched review REV-7f70c161 (decision *accept*) covers formal correctness only. Source commit `3ecf1f474ae3868e877727d1061c444c5307271f`, `lake build` 3167 jobs, 0 warnings (build evidence `research/ecdlp/runs/RUN-62dbda7c-fe4f-4432-84a2-2068cd516355-command-3aa1a646-4c3d-4ffa-b511-6c3b1ba10f31.evidence`); axioms `[propext, Classical.choice, Quot.sound]` (`research/ecdlp/runs/RUN-62dbda7c-fe4f-4432-84a2-2068cd516355-command-ec79161e-5786-4124-9252-7d36a4adcc4e.evidence`). In `MathResearch/ECDLP/RationalSLP.lean` the operation type `Op` additionally has a division `div i j`; values are evaluated in F (with Lean's total convention $x/0 = 0$), and `Defined prog u` says that *every* division of `prog` (including dead ones) has a nonzero divisor value at u. The statement, verbatim from the repository (namespace `MathResearch.ECDLP.RationalSLP`; the binders `{F : Type*} [Field F] [DecidableEq F]` come from a `variable` line):

```
theorem length_bound_rational_slp {W : Affine F} (h1 : W.a₁ = 0) (h2 : W.a₂ = 0)
    (h3 : W.a₃ = 0) (h2' : (2 : F) ≠ 0) (P : W.Point) (r : ℕ) (hr : addOrderOf P = r)
    (N : ℕ) (hN : 2 * N < r) (prog : List (Op F)) (i : ℕ)
    (hdef : ∀ k : ℕ, 1 ≤ k → k ≤ N → Defined prog (xCoord (k • P)))
    (hprog : ∀ k : ℕ, 1 ≤ k → k ≤ N → (evalAt prog (xCoord (k • P))).getD i 0 = (k
    : F)) :
    N ≤ 3 * 2 ^ prog.length + 2
```

In words: for a straight-line program over F with $+, -, \times, \div$ and constants and a single input u, if every division has a nonzero divisor at $u = x(kP)$ and the i-th output equals k there for all $1 \le k \le N$, $2N < r$, then $N \le 3 \cdot 2^L + 2$ where L is the program length (so $L \ge \log_2((N-2)/3)$). The constant is the same as in the division-free case (i). The proof

(as recorded) obtains from the program a pair of polynomials (A, B) of degree $\leq 2^L$ with $A(x(kP)) = k\,B(x(kP))$ and $B(x(kP)) \neq 0$, and applies a rational analogue of Theorem C (`degree_bound_rational_short_weierstrass`: $N \leq 3d + 2$ for $\max(\deg A, \deg B) \leq d$, no coprimality hypothesis assumed); the non-vanishing of the cleared polynomial uses orbit propagation, which forces $2 = 0$ in F. The review REV-7f70c161 records that the bound $3d + 2$ is attained at $d = 1$ ($p = 11$, $y^2 = x^3 + 1$, $P = (9, 2)$, $A = 1 + u$, $B = 3 + 2u$, $N = 5$). The Lean statement for odd r in the form $r \leq 6 \cdot 2^L + 5$ is `length_bound_rational_slp_order` in the same file (not separately recorded in the ledger records we consulted). *Scope.* The model has no branching or comparison, no bit access, and does not take P as input; it counts field operations only and makes no bit-complexity claim. It does not apply to rho, baby-step giant-step or index calculus. The ledger's earlier remark that the true constant might be nearer 2 is refuted as a uniform claim by the attained instance above.

(iii) Theorem DT: algebraic computation trees with equality tests; formally verified, proof reviewed. Ledger record HYP-dbe79a50 (status *open*, proof status *formally verified*, outcome *verified*), Lean record PRF-ab954d65; the blind adversarial review REV-d7bc5940 confirmed the paper proof of Theorem DT and independently found the gap in the paper proof of its Lemma A that the Lean proof repairs (propagation from one adjacent pair in both directions). Source commit `b699c3e8ac3ba3d96b1f6b6bc0fcf7b3eea34cbd`, `lake build` 3168 jobs, 0 warnings (build evidence `research/ecdlp/runs/RUN-62dbda7c-fe4f-4432-84a2-2068cd516355-command-9ccbf89a-509f-496d-b41f-a352be9c57c5.evidence`); axioms `[propext, Classical.choice, Quot.sound]` (`research/ecdlp/runs/RUN-62dbda7c-fe4f-4432-84a2-2068cd516355-command-c8fc5d7e-6598-4dbb-81dc-f3bc94eeb0f2.evidence`). Lean source `MathResearch/ECDLP/DecisionTree.lean`. The Lean statement, verbatim (namespace `MathResearch.ECDLP.DecisionTree`; same binders as above):

```
theorem Tree.length_bound {W : Affine F} (h1 : W.a₁ = 0) (h2 : W.a₂ = 0)
    (h3 : W.a₃ = 0) (h2' : (2 : F) ≠ 0) (P : W.Point) (r : ℕ) (hr : addOrderOf P = r)
    (N : ℕ) (hN : 2 * N < r) (T : Tree F) (L : ℕ) (hL : T.arithDepth ≤ L)
    (hrun : ∀ k : ℕ, 1 ≤ k → k ≤ N → T.exec (xCoord (k • P)) = some (k : F)) :
    N ≤ 9 * 2 ^ L - 1
```

The informal statement recorded in the ledger (verbatim from the HYP record): *Theorem DT (algebraic decision trees with equality tests): over a field F with $2 \neq 0$, W short Weierstrass, P of order r, $2N < r$. Any algebraic computation tree with single input $u = x(Q)$, arithmetic nodes $+, -, \times, \div$ (divisions defined where executed) and branching only on "register $= 0$?", in which every root-to-leaf path has at most L arithmetic nodes, and which outputs k at $u = x(kP)$ for all $1 \leq k \leq N$, satisfies $N \leq 9 \cdot 2^L - 1$ regardless of depth.* The formal model: a `Tree` is a leaf (outputting a register), an arithmetic node (an `Op`: `const`, `var`, `add`, `sub`, `mul`, `div`), or a test "register $= 0$?" with two subtrees; register 0 is the input; `arithDepth` is the maximal number of arithmetic nodes on a root-to-leaf path and tests are not counted; `exec` returns `none` if an executed division has a zero divisor.

Scope (after the complexity review REV-6c73a098 of remarks E0 and E3). Theorem DT is a statement about *equality branching only*: the only tests are "register $= 0$". *Constants are charged*: in the formal model every `const` node is an arithmetic node and counts toward L (the informal text above does not say this; the ledger note and the review do). With *order comparisons* the arithmetic length becomes vacuous: the review records that a binary-search tree over the sorted values has 0 arithmetic nodes (toy $N = 300$: depth 9, 300 leaves), so in a model with comparisons only tree size and depth matter, and no size–depth lower bound is proved. Theorem DT therefore does *not* bound algorithms that compare integers or inspect bits of field elements, and it bounds arithmetic nodes, not tree size, depth or running time. The remarks about Shoup's generic group model recorded with Frontier E (E0) are reviewed only as heuristics and are not used here. The ledger also records Theorem P and Corollary P1 for

this model; they are *not* formalized, Corollary P1 was false as transcribed (missing hypothesis $\operatorname{char} F = 0$ or $\operatorname{char} F > N$; counterexample over $\mathbb{F}_{3^7}$) and is superseded by HYP-8e55674a; we do not state them. Novelty of Theorem DT is unchecked (the ledger notes Ben-Or/Strassen-type degree arguments and Lange–Winterhof as possibly related).

(iv) Method barrier L2: proof attempt, PI-checked, not reviewed. Ledger record HYP-0d4f14da (status *open*, proof status *proof attempt*; the principal investigator checked the proof steps and reproduced the supporting numerical scripts; not formalized, not independently reviewed). The claim, as recorded: for a "translation annihilator" $D = \sum_{i \in S} c_i G(Q + iP)$ with $\sum c_i = \sum i c_i = 0$ ($|S| = s$, width w), $G = g \circ x$, $\deg g = d \geq 1$, $2N < r$, if $D \not\equiv 0$ then D has pole degree exactly $2ds$ and at least $2(N - w)$ zeros, so $N \leq ds + w$, minimized at $s = 3$, $w = 2$, which is Theorem C. The zeros arise at the points $Q = kP$ where all translates are in the prescribed range: there $D(kP) = \sum_i c_i f(k + i) = 0$ because the prescribed values $f(k) = k$ have degree ≤ 1 and $\sum c_i = \sum i c_i = 0$ (we spell this out; the ledger states only the zero count). Hence zero-counting arguments with translates cannot improve the constant $1/3$; any sharper bound must use information beyond the pole structure and the values $G(kP) = f(k)$. We state this only as recorded; it has no proof in this report. The same record contains a claim L3 on anomalous $j = 0$ curves which we do not use.

(v) Empirical conjecture C* (not proved). Ledger record HYP-a8d1f541 ("Conjecture C*", status *open*, experimental): outside structured families (for example anomalous CM $j = 0$ curves), the minimal degree of an interpolating g is $N - 1 - O(1)$. The ledger records 154,288 toy cases over prime fields with $p \leq 997$ and a maximal observed deficit of 3. This is an observation at toy sizes, not evidence for large parameters. If true it would show that the factor 3 between the bounds above and the trivial Lagrange upper bound is a limitation of the method, not of the problem.

(vi) Theorem E (cusp-degeneration bound): reviewed proof, *not formalized, not formally verified.* Ledger record HYP-a4caa1de ("Theorem E (corrected after review)", status *completed*, proof status *reviewed*, outcome *verified*; it supersedes HYP-38005bec, where the same statement appeared under the label "Theorem D"). One blind review, REV-3b9427bd (decision: revise, i.e. accept with three statement corrections; the corrected statement is the one below), examined it by symbolic recomputation for $N = 3, 4$ and exhaustive small-p tests; the proof is a paper proof using the Schwartz–Zippel lemma (listed in the ledger as an assumption) and the standard theory of division polynomials, taken as standard and not re-proved. No Lean formalization exists, no human has checked it, and it is *not* a formally verified result. The ledger claim, verbatim from the record:

> (Theorem D, corrected) Let $N \geq 3$, $p > 2N$ prime, $V := \{\tau \in \mathbb{F}_p^3 : 4A^3 + 27B^2 \neq 0,\ W(\tau) \neq 0\}$ (the valid triples, on which mindeg_N is defined and equals deg of the interpolant), $Z := \{\tau \in V : c_{N-1}(\tau) = 0\}$. If p does not divide $\operatorname{Num}(\tau_0)$ (always true for target k^2; for target k iff $p \notin \operatorname{Bad}_N$), then $|Z| \leq w_N p^2$, hence $\operatorname{mindeg}(N) = N - 1$ for all $\tau \in V \setminus Z$. [$|\mathbb{F}_p^3 \setminus V|$ is separately bounded by $(6 + \deg W)p^2$ and is not covered by this theorem.]

Notation (from the review report `research/ecdlp/runs/RUN-d55e8925-6e2d-41b4-8548-10a111df3297.md`, which also gives the corrected statement): $\tau = (A, x_0, y_0)$ with $B := y_0^2 - x_0^3 - Ax_0$ (the curve is $y^2 = x^3 + Ax + B$ and the point is (x_0, y_0)); ψ_m are the division polynomials; $W := \prod_{1 \leq a < b \leq N} \psi_{a+b}\psi_{b-a}$; $c_{N-1}(\tau)$ is the divided difference of order $N-1$ of the target function (k^2 or k) at the nodes $x(kP)$, $1 \leq k \leq N$, so that c_{N-1} is the leading coefficient of the interpolant; $\operatorname{Num} := W\, c_{N-1}$ as an explicit polynomial; $\tau_0 = (A, x_0, y_0) = (0, 1, 1)$ is the cuspidal point; $\operatorname{Bad}_N := \{p > 2N \text{ prime} : p \mid \operatorname{Num}(\tau_0)\}$ (a finite set of primes, the primes $> 2N$ dividing the numerator of $c_{N-1}(0)$ for target k; the review lists, e.g., $\operatorname{Bad}_3 = \{11\}$ and $\operatorname{Bad}_4 = \{151\}$); w_N is half the weighted degree of Num (a bound for its total degree after substituting B). The

review's values are $w_3 = 23$ and $w_4 = 81$. *Scope.* The theorem concerns fixed small N: the bound $w_N p^2$ is non-vacuous only for p large compared with N (the review notes $N \lesssim (3p)^{1/4}$ for non-vacuity); in the exhaustive tests it was genuinely tested only for $N = 3$, $p \in \{29, \ldots, 47\}$. The ledger records that it does not prove Conjecture C* beyond the regime $N < p^{1/4}$, that it is not an algorithm, and that it gives no complexity claim.

8 Benchmarked algorithm candidates

This section is *not* a theorem section. It summarizes the ledger records of rounds 11–13 of the laboratory whose recorded outcome is *refuted* (here: the candidate was tested and found not to beat Pollard rho, or the tested signal was null). Every row is an *experiment* at toy sizes (prime-field curves of at most about 44 bits, one case of $p = 524309$), with a toy Python implementation; the ledger notes for each of these records that no independent review was done (HYP-6da71353 additionally contains two unreviewed derivations, discussed in Section 7(iii)). Each record is proof status *experimental*. None of these is a proof that no better algorithm exists, and none establishes any property for large parameters. All numbers below are copied from the ledger records named in the first column; we have not recomputed them. "rho" is Pollard rho on the same toy instances unless noted. **No improvement over Pollard rho is claimed, and none was found.**

Record	Mechanism	Measured scaling / cost ratio vs rho	Exact obstacle
HYP-23351e75 Inventor A (ECDLP-42, lifted relations)	A1: Farey-height factor-base index calculus with meet-in-the-middle relation search; A2: LLL (Coppersmith-style) decomposition into two Farey points; A3: lift to $E(\mathbb{Q})$ on $y^2 = x^3 - 2$.	A1: 66/66 correct (12–24 bits); cost vs rho at 24 bits 42.7× (4-term), 28.9× (6-term); 3-term control $\sim 260\times$ at 20 bits; measured exponents 0.82/0.71 vs model 0.667/0.60 (rough); pre-registered verdict NO-ADVANTAGE. A2: 0 of 124/77/61 existing decompositions found at 16/20/24 bits. A3: planted 50/50, random 0/300 (coverage 10^{-3} to 4×10^{-7}).	A1: with s unknown logs, hashing needs $\sim \sqrt{ps}$ additions $> \sqrt{p}$. A2: rank-one structure is not a lattice condition. A3: random coverage 10^{-3} to 4×10^{-7}.

Record	Mechanism	Measured scaling / cost ratio vs rho	Exact obstacle
HYP-d6cc9bce Inventor B (ECDLP-43, representation change)	B1: automorphism-orbit rho ($j = 0$, $\lvert\mathrm{Aut}\rvert = 6$; $j = 1728$, $\lvert\mathrm{Aut}\rvert = 4$); B2: 2-isogeny volcano ascent to $j = 0$ then B1.	B1: 450/450 runs, $r \leq 2^{36}$; fitted exponents 0.44–0.49 (CIs contain or nearly contain 0.5); pooled rho/orbit ratio 1.45 $[1.14, 1.88]$ ($j = 0$, below the pre-registered 1.15 threshold as recorded) and 2.62 $[2.15, 3.17]$ ($j = 1728$, inflated by a heavy-tailed rho baseline 2–3× above textbook). B2: 150/150; isogeny evaluations $\sim a^{1.86}$ (polylog); the pre-registered rule printed "asymptotic improvement" but this is a false positive from additive overhead shrinking 60% → 3%; medians indicate exponent 1/2, constant ∼ 2–2.5.	Known technique (constant $\sqrt{\lvert\mathrm{Aut}\rvert}$). B2 applies only to curves isogenous to $j = 0$ (fraction $\sim p^{-0.47}$ $[-0.58, -0.38]$); equivalent in kind to Jao–Miller–Venkatesan / Galbraith volcano climbing. Isogeny-pooled, twist-pooled and other curve models rejected on paper (they preserve r and group structure).
HYP-d913fa61 Inventor C (ECDLP-44, beyond-rho search)	C1: negation-class rho; C2: preprocessed fixed-walk table; C3: multi-instance shared distinguished-point (DP) search.	Reference DP rho (31-way) exponent 0.486 $(0.467, 0.506)$, 20–44-bit toy curves. C1 (frozen) failed: undetected fruitless cycles (period ≥ 6), $\sim 11.8\sqrt{n}$ at 40 bits, exponent 0.687, 44-bit censored; post-hoc C1b exponent 0.482, pooled gain ∼ 1.13 (CI ∼ 0.95–1.33) $< \sqrt{2}$, constant-factor rule not met. C2: online exponent 0.344 $(0.317, 0.371)$, preprocessing $\sim n^{0.62}$, $ST^2/n \sim 4$, break-even 15–71 queries. C3: per-instance cost $\sim M^{-0.57}$ $(-0.63, -0.50)$, $\sqrt{M}$ law; n-exponent unchanged 0.465; gain 1.7–3.1× at $M = 16$.	No single-instance exponent below 1/2. C2 lies on the generic $ST^2 \sim N$ tradeoff curve, not beyond it. C3 gain is the known $\sqrt{M}$ law. Ledger: all known techniques, no novelty.

Record	Mechanism	Measured scaling / cost ratio vs rho	Exact obstacle
HYP-720e359e Inventor D (ECDLP-47, algebraic relation finder)	Index calculus with S_3 decomposition over a Petit–Kosters–Messeng multiplicative-coset factor base ($x \in g\mu_d$, $d = 2^n \mid p-1$), relation finding by Gröbner/enumeration.	65/65 runs recovered k (1581 sampled relations verified; sympy Gröbner matches exhaustive search for p up to 17921); total cost exponent ~ 1 (0.99 $[0.92, 1.06]$) vs rho ~ 0.5; D2a/rho = 11 (16 bits), 94 (20), 633 (24), 2533 (28, post hoc); D2b ($d \sim p^{1/3}$) exponent 1.075; linear algebra $< 0.1\%$.	Per-target finder cost $\sim \lvert V\rvert$, relation $\sim 2p/d$ additions, total $\sim p$ independent of d. Hash/enumeration finders cost $\geq p^{1/2-1/(2m)}$ per relation while the accounting needs $T < p^{1/2-1/m}$; Gröbner/resultant collapses to a degree-$\lvert V\rvert$ univariate problem (time $\sim d^{2.1}$–d^3); no subfield over $\mathbb{F}_p$ for Weil restriction.
HYP-32e353ac ETH-technique transfer (ECDLP-46; source "openai/math family 378", external, not peer reviewed)	Transfer of an ETH-refutation technique to ECDLP: direct CNF (SAT) route; summation-polynomial decomposition; predicates on x alone; output bits of $x(kP)$.	Negative on every route, no prototype (decision gate not met). Direct CNF: affine encoding $\sim 48\,\ell^{2.83}$ variables, ladder $\sim 503\,\ell^{2.41}$; paper parameters (Lemma 4.8, $\beta = 1$) only become non-vacuous at $\log_2 n \sim 10^{10}$; summation-polynomial route costs $p^{2/(m+1)}$ total even with a free oracle; S_3 test encoding $\sim 45.5(\log_2 p)^2$ variables; ANF density of output bits ~ 0.5 up to degree ℓ.	Time $2^{Cn/\sqrt{\ln n}}$ beats rho only if $n/\ell < \sqrt{\ln n}/(2C)$, impossible for $n = c\,\ell^a$, $a > 1$ (extrapolation from $\ell \leq 16$ is crude, but the ledger records the crossover inequality as exact); predicates on x alone are Q-independent; no sparse ANF form. Parameter thresholds depend on the agent's reading of Sections 3–5 of the source (unreviewed).
HYP-93fbd7d0 Frontier exploration D (ECDLP-37, follow-ups)	(1) Structure-aware SAT encodings (x-only Montgomery ladder; native XOR via CryptoMiniSat); (2) Hensel-section cocycle $g(kP) = k\,g(P) - NS_k$ with Birkhoff sum S_k; (3) lift-variation idea.	No cost ratio vs rho is recorded. (1) Affine conflict exponents 1.27 (Glucose4), 1.47 (CaDiCaL); the ladder was unsolved within 120 s at $n = 5$–7. (2) Identity verified on 8 curves ($p \leq 30011$); no low linear complexity, P-recursive or low-degree relation and no $k \bmod m$ dependence in the p^2/p^3 digits (the spectral flag was a null-model artefact; symmetry-respecting null $\lvert z\rvert \leq 2.6$, post hoc). All NULL.	(1) Structure-aware encodings are worse than the affine encoding. (2) The identity holds but the digits show no exploitable structure. (3) Killed: 4 unknowns vs 3 equations; the lift-dependent part is not low-degree. Not reproduced by the PI (long SAT runs); CryptoMiniSat `key_ok` flags invalid in some rows (indexing bug fixed, 9 instances re-validated by the agent).

Record	Mechanism	Measured scaling / cost ratio vs rho	Exact obstacle
HYP-6da71353 Frontier exploration E (ECDLP-41, integer structure)	Distinguishers on integer structure of $x(Q)$ at $p = 524309$ (two prime-order curves, seeds $1, 2$): E1 local carry/bit/order/difference correlations of $x(Q)$ vs $x(Q + jP)$; E2 index differences of integer-sorted neighbours; E6 three-term carry $c_k = [x_{k+1} + x_{k-1} \geq p]$; E4 HNP-lattice; E5 lifting smoothness.	No cost ratio vs rho or scaling exponent is recorded. E1 null (a 2% planted signal was detected, $z = 28$–760); E2 null; E6 fails the pre-registered two-seed rule ($p_T = 0.025$, 0.050 at different h; positive control $p_T = 0.005$); E4 and E5 killed by derivation/literature.	E5: $\sim 2^{18}$-bit numerators at rank 29 for a 256-bit p. Toy size only; absence of signal is not absence of structure. E1 null draws were reduced (60/20) after a timeout (recorded in an amendment).

Reading the table. In rows A and D the measured total cost is above rho's at the tested sizes (ratios in the table). Rows B and C report constant-factor effects of known techniques; the fitted exponents recorded for them are 0.44–0.49 (B1) and 0.482–0.486, 0.465 (C), and the only recorded exponent below 1/2 is the *online* exponent 0.344 of the preprocessed table C2, whose preprocessing costs $\sim n^{0.62}$ (a different cost measure, on the generic tradeoff curve). The ETH-transfer and Frontier rows are model computations and null structure searches without a cost ratio against rho. The ledger outcome *refuted* refers to the hypothesis that the candidate beats rho (or carries a usable signal), at the tested sizes. This table is the laboratory's measurement record, not a survey of the literature on ECDLP algorithms, and it makes no novelty claim for the known techniques it re-measured.

9 Limitations

- Degree is not time. A lower bound on $\deg g$ for polynomials in one variable $x(kP)$ does not bound the running time of any algorithm, and we draw no bit-complexity consequence.
- The results concern interpolation from x-coordinates only, with exact agreement on a range (B, C) or a subset (D); they say nothing about approximate or probabilistic computation or about other coordinates.
- Existence and tightness: interpolating polynomials of degree $\leq N - 1$ exist by Lagrange interpolation, so the bounds are tight up to a constant factor; we prove no exact value of the minimal degree (see the conjectural remark in Section 7).
- The ledger's rational-function versions (Theorem A, part (1) of the partial-agreement hypothesis) and the density corollary are not formalized.
- Theorem D's extra hypothesis is removed over $\mathbb{F}_p$ only by an informal Hasse-bound argument.
- The straight-line-program bounds of Section 7(i),(ii) are for restricted, branch-free, single-input models (with and without division); the bit-cost corollary is not formalized. Theorem DT covers equality branching only, charges constants, and is vacuous for arithmetic length once order comparisons are allowed. None of these models contains rho, baby-step giant-step or index calculus.

- The method barrier (iv) is an unreviewed proof attempt, the conjecture C* (v) is unproved, and Theorem E (vi) is a reviewed paper proof that is *not* formalized and was tested only at $N = 3, 4$.
- The benchmark table of Section 8 consists of toy-size experiments without independent review; its numbers are copied from the ledger and were not recomputed for this report.
- No improvement over Pollard rho is claimed.
- The universal $O(n \log n)$ ECDLP target remains open; nothing here improves on any known ECDLP complexity.
- Novelty is unsettled (Section 6); no human has verified the mathematics, the Lean-to-statement correspondence, or the bibliography.

Provenance note. This report was produced by an AI research laboratory (this draft and its revisions 1 and 2 by worker agents reporting model identity Sonnet 5.5, `claude-sonnet-5-5`, not verifiable from runtime metadata) and reviewed by AI agents; no human verification has yet taken place. It is an internal technical report and not a submission.

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